VINAYAK GUPTA

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Professional Summary

Backend and Systems Engineer with 3 years of experience specializing in high-concurrency Go architectures, edge AI optimization, and low-level data engineering . Proven track record deploying real-time YOLO computer vision pipelines on NVIDIA Jetson infrastructure, building high-performance cryptographic Key Management Systems (KMS) in Go with sub-2ms latency, and managing complex integrations with legacy national APIs . Expert at bridging the gap between advanced AI tooling, firmware/PLC automation layers, and rugged, production-ready backend systems.

Technical Skills

Languages	:	Go, Python, TypeScript, Rust, OCaml, JavaScript, C++, SQL, Java
AI & Systems	:	NVIDIA Jetson Nano, RTX 3090, TensorRT, YOLO, OpenCV, Modbus TCP, LLM Code Gen (Gemini)
Backend & Security	:	Node.js (Express/TS), Django, Echo (Go), REST APIs, Custom KMS, AES-GCM, PBKDF2
Tools & DevOps	:	Docker, PostgreSQL, MongoDB, Git, Linux (Arch/Nvim customization), CI/CD, Tauri

Experience

Techasoft Pvt. Ltd. — Bangalore, India
Software Development Intern (Edge AI & Automation) — Mar 2026 – Present

- Architected and validated real-time edge computer vision pipelines for industrial FMCG production, deploying CNN YOLO models onto NVIDIA Jetson Nano to process dual camera feeds at 30 FPS.
- Engineered hardware-software control loops via Python (Modbus TCP), translating CV inference into low-latency 24V PLC signals within an 8ms window for automated pneumatic packaging rejection.
- Designed a pixel-to-millimeter ROI calibration system in Django, dynamically calculating geometric coordinate matrices based on factory pixel scales (px/mm) to eliminate manual labor.
- Optimized high-concurrency data layers by migrating production systems from SQLite to PostgreSQL, enhancing indexing strategies for on-site real-time data logging and parameter thresholding.

Medoc Health IT — Remote
Engineering Product Lead & Project Management Intern — Aug 2023 – Jul 2025

- Developed a high-throughput, low-latency Key Management System (KMS) in Go using PBKDF2 for master key derivation, achieving sub-2ms key-retrieval latency via the Echo framework.
- Wrangled legacy national healthcare infrastructure by engineering asynchronous backend integrations for the ABHA API ecosystem, reverse-engineering undocumented endpoints for 40,000+ patients .
- Architected the MedocEUA full-stack platform (Node.TS/MongoDB), engineering an optimized $2^n - 1$ priority triage algorithm for deterministic emergency queuing and decentralized consent management .
- Directed delivery for medical telemetry modules using Flutter, implementing bi-directional digital prescription workflows and background data sync via Android Health Connect API .

Medoc Health IT — Remote
Software Development Intern — Aug 2022 – Jul 2023

- Built secure backend REST APIs and containerized workflows utilizing Node.js, Docker, and SQL, implementing robust server-side validation and CI/CD automation pipelines .

Projects

Toy-Lang LLVM Compiler — Code
OCaml, Rust, LLVM IR, Gemini Antigravity

- Designed a strongly-typed functional language with a strict affine type system (borrow checker), tagged unions, and explicit module-name resolution patterns (e.g., *Std.Math.Trig*).
- Engineered an automated tree-shaking compilation pass leveraging Gemini Antigravity to formalize grammar and prune dead code, emitting optimized, minimal-signature LLVM Intermediate Representation.
- Built compile-time safety structures featuring absolute side-effect isolation for functions vs. unity-returning procedures and deterministic null-type isolation mechanics.

Zundrath Key Management System — Code
Go, Cryptography, Echo, PBKDF2

- Developed a local cryptographic microservice serving concurrent clients with sub-2ms retrieval latency via mutex-protected on-demand disk indexing .
- Implemented a fault-tolerant, two-phase key rotation pipeline providing a 24-hour safe-migration window to eliminate decryption blackouts during live application schema updates.

Grounded Object Data (GOD) Format — Code
Go, EBNF Grammar, Lexical Scanner, Reflection API

- Designed and engineered a lightweight, schema-less data serialization format utilizing a strict EBNF grammar specification to eliminate JSON payload redundancy and reduce data transmission footprints.
- Implemented a hand-written lexical scanner and recursive-descent parser in Go, introducing native, first-class tabular structures (`header:rows`) that strip repetitive object keys from multi-row payloads.
- Enforced strict "grounded" type-safety semantics within the marshalling engine, structurally preventing null-pointer runtime panics by automatically initializing missing fields to zero-values or explicit `\0` null-terminators.

Education

B.Tech in Computer Science and Engineering — 2021-2025
LNCT University, Bhopal